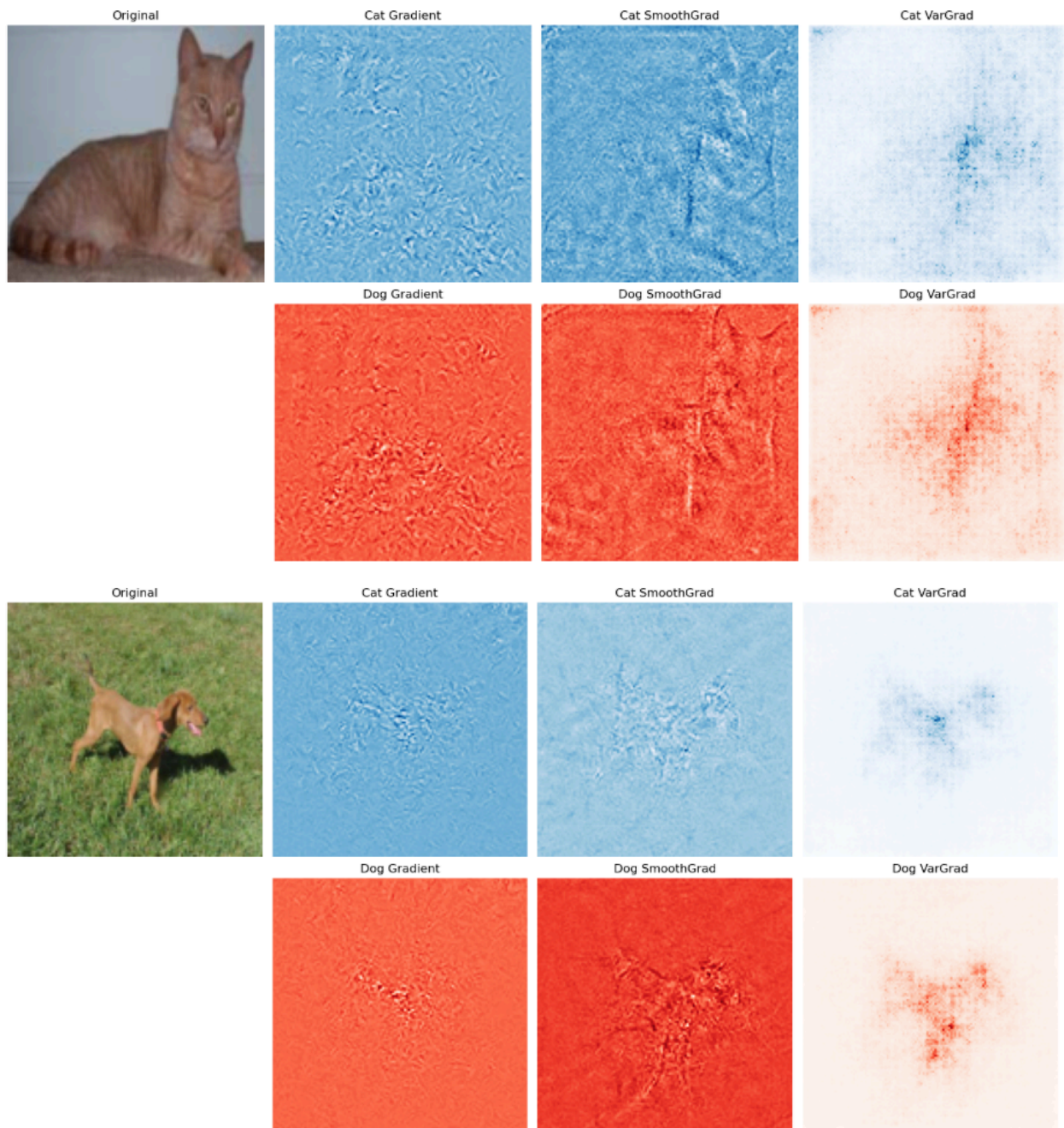
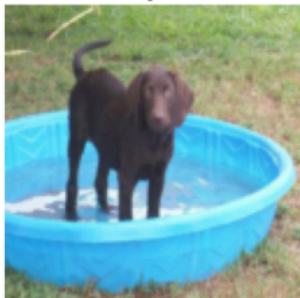


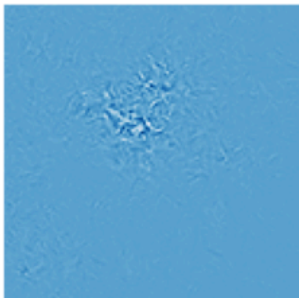
2.



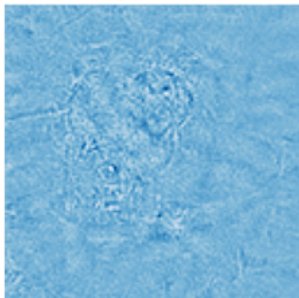
Original



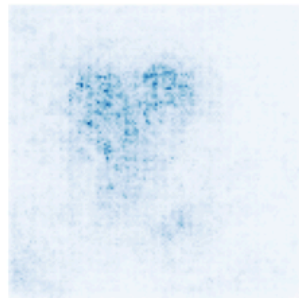
Cat Gradient



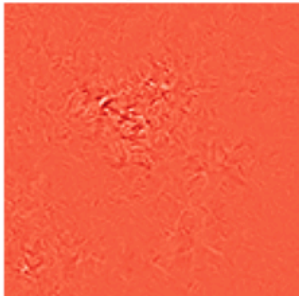
Cat SmoothGrad



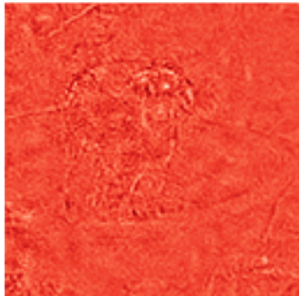
Cat VarGrad



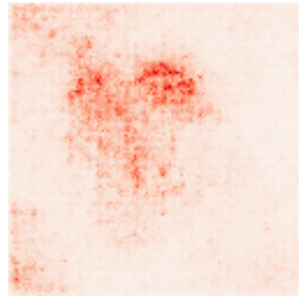
Dog Gradient



Dog SmoothGrad



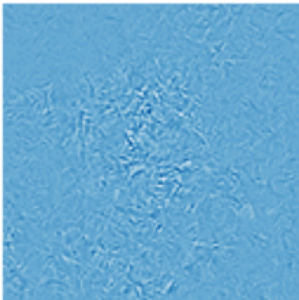
Dog VarGrad



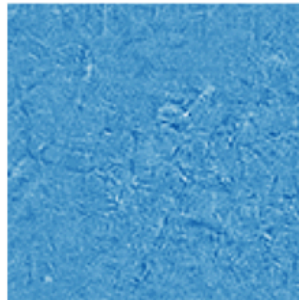
Original



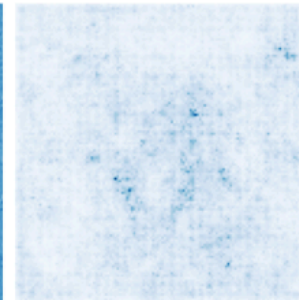
Cat Gradient



Cat SmoothGrad



Cat VarGrad



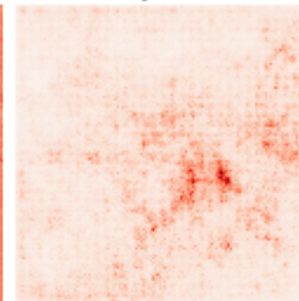
Dog Gradient

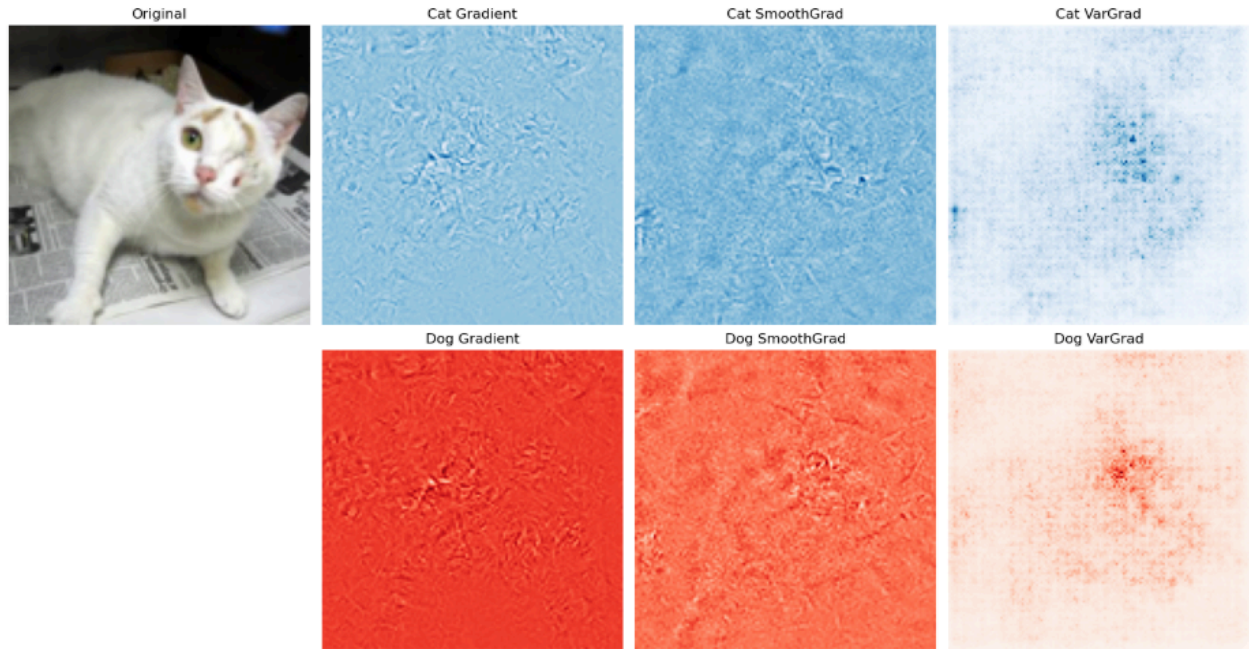


Dog SmoothGrad



Dog VarGrad

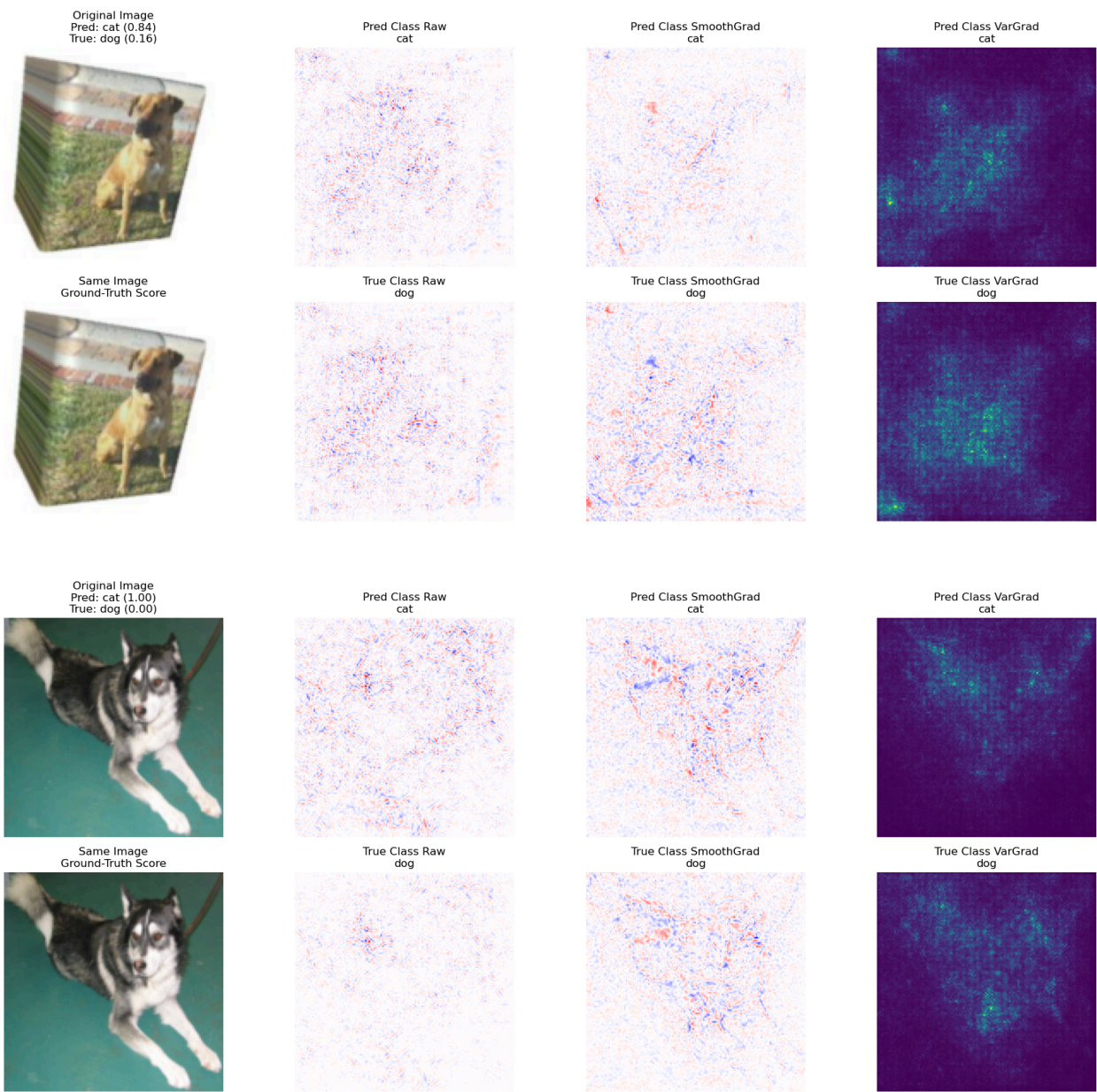




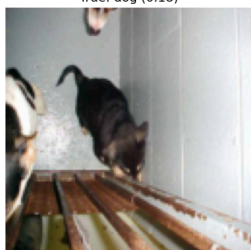
These images show the differences in the different methods, from them we can see that the standard gradient is very noisy and pixelated with minor highlights of the cat/ dog. It is difficult to see and understand so it is not good for high level features. The smooth grad does a good job at making the features more discernable and reducing noise, by averaging the gradients it makes the outlines much cleaner. The vargrad uses the variance of the gradient and reduces noise to a more contrasting color so the focal point of the class score is easier to distinguish.

Both smoothgrad and vargrad show the silhouette of the animal which can be a little lost in the standard gradient. Smoothgrad does provide a bit better of a highlighted feature map as the shape class is easier to make out even if the colors are more the same. The standard gradient suffers the most from high frequency noise than the smoothgrad and vargrad because changing the individual pixel value has large changes in the output, vargrad and smoothgrad act as better filters to the noise. The localization on the smoothgrad and vargrad is much better as it is able to focus more on the animal and less on the grass in most images.

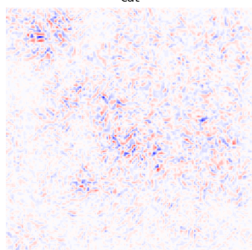
Misclassified animals:



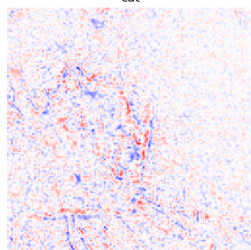
Original Image
Pred: cat (0.82)
True: dog (0.18)



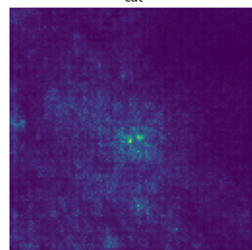
Pred Class Raw
cat



Pred Class SmoothGrad
cat



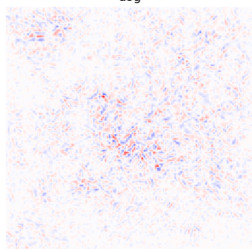
Pred Class VarGrad
cat



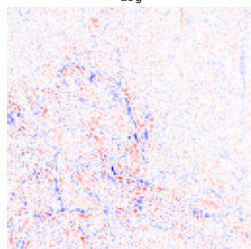
Same Image
Ground-Truth Score



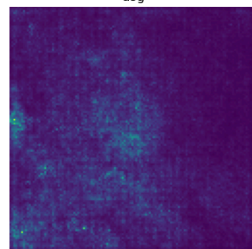
True Class Raw
dog



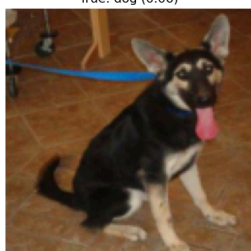
True Class SmoothGrad
dog



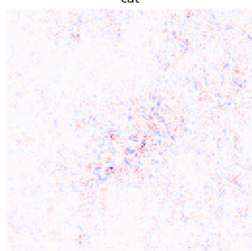
True Class VarGrad
dog



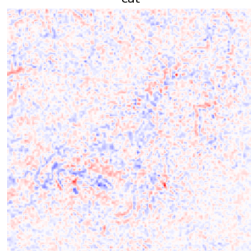
Original Image
Pred: cat (0.94)
True: dog (0.06)



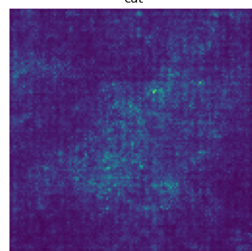
Pred Class Raw
cat



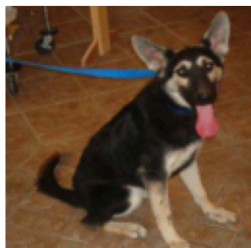
Pred Class SmoothGrad
cat



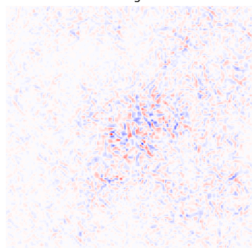
Pred Class VarGrad
cat



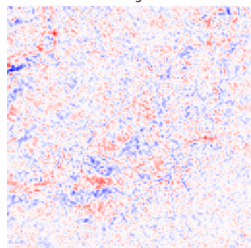
Same Image
Ground-Truth Score



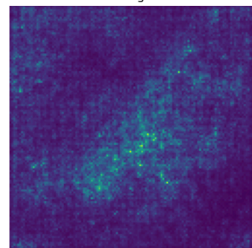
True Class Raw
dog

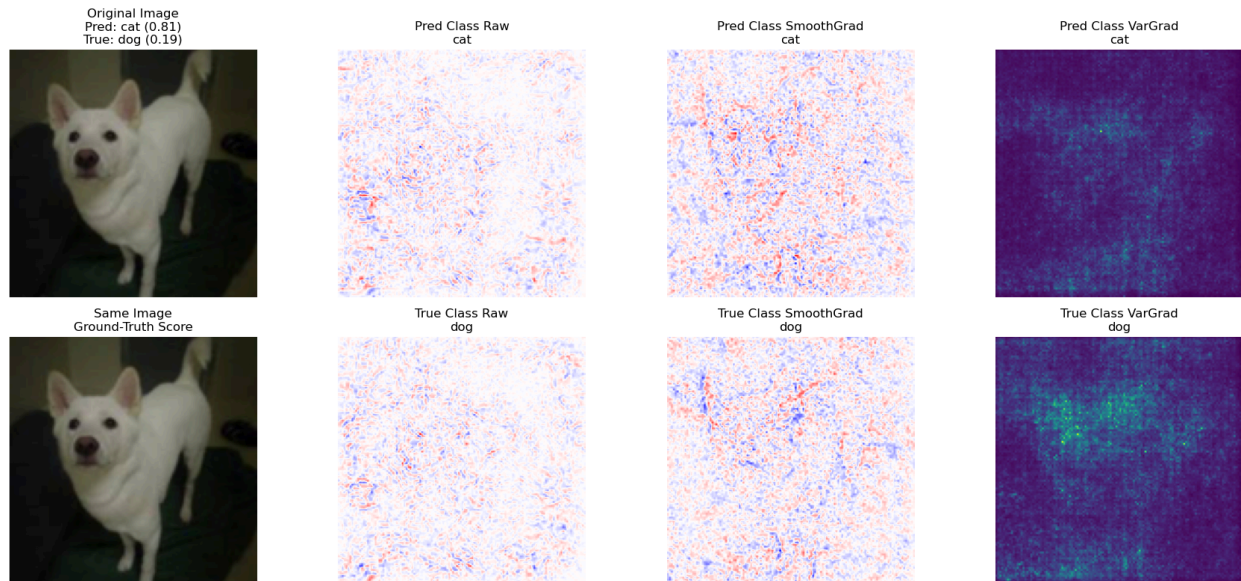


True Class SmoothGrad
dog



True Class VarGrad
dog





These are the different scalar outputs for the misclassified images, these show what the system used to see what was a dog and what was a cat. After looking at the vargrad I can see what was considered an unstable region and that it most of the time can be seen that the dog classifier is lit up more so to understand the missclassification we have to look at the standard gradient, in this we can see that some of the maps the cat classifier is lit up a bit more whether to random noise or something else.

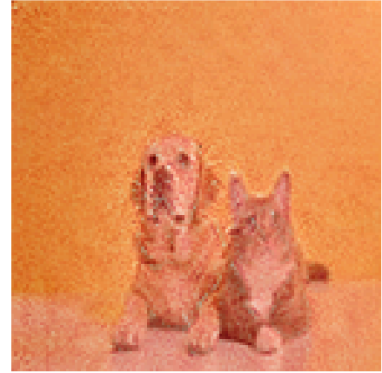
Original Image
Pred: dog (0.91)



Source URL: <https://en.photo-ac.com/photo/28912931/dog-and-cat>
Cat-Class SmoothGrad
Score: 0.093



Dog-Class SmoothGrad
Score: 0.907



Source URL: <https://en.photo-ac.com/photo/28912931/dog-and-cat>

Original Image
Pred: dog (0.91)



Combined Overlay
Blue = cat-class, Red = dog-class
Suppression rule: |value| < 0.3 set to 0



Original Image
Pred: dog (0.67)



Source URL: <https://stock.adobe.com/search?k=cats+and+dogs>
Cat-Class SmoothGrad
Score: 0.331



Dog-Class SmoothGrad
Score: 0.669

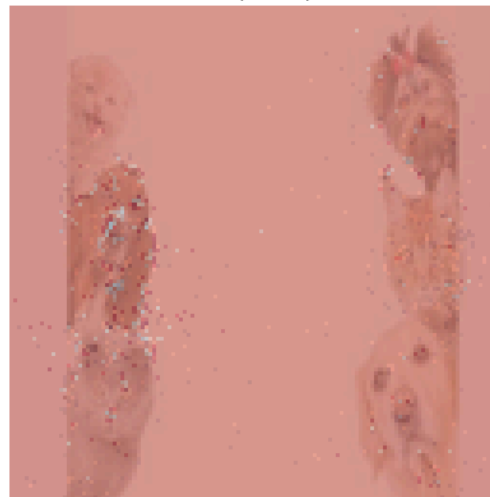


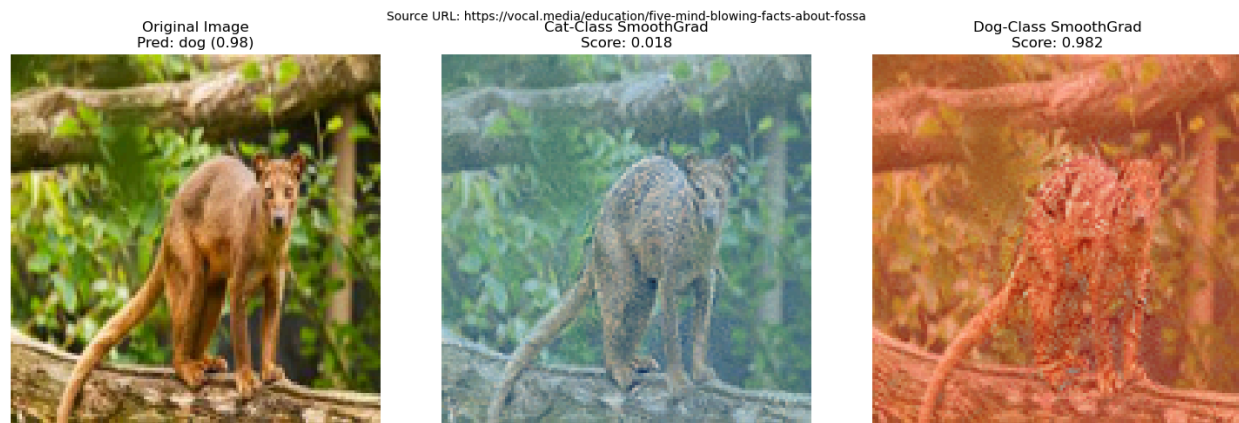
Source URL: <https://stock.adobe.com/search?k=cats+and+dogs>

Original Image
Pred: dog (0.67)



Combined Overlay
Blue = cat-class, Red = dog-class
Suppression rule: |value| < 0.3 set to 0





Source URL: <https://vocal.media/education/five-mind-blowing-facts-about-fossa>



In the first image we can see that because the faces of the cat and dog are close together the score is more focusing on the predominant feature and not being activated as much in the cat layer. In the overlay it shows that the dogs face is much more recognized than the cat.

In the second image that shows the multiple cats and dogs we can see that the model is not able to focus on all of the objects and is focusing on the middle. It is assumed that the long droopy ear features are more overpowering than the cat face but it is still focusing on the cats and recognizes them which is probably why the class score is so low.

I chose a fossa because it has features of both cats and dogs and I wanted to see how the features would be broken down, it seems like the model has identified it as mostly a dog with little difficulty seemingly due to its abdomen and legs, I thought it would look at its ears and face more and be a bit more confused. This makes me think that the model can choose to focus on specific features that are more definite and discriminate against other features that might go against the classification.

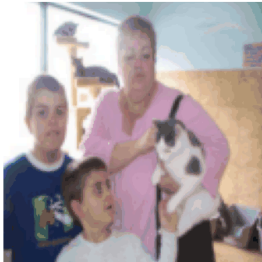
Selected Cat Image



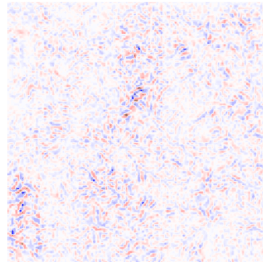
Selected Dog Image



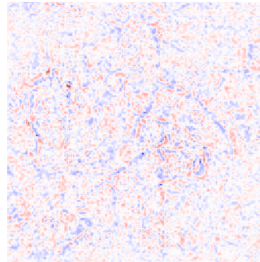
interpolated image
t = 0.00
Pred: dog (0.86)



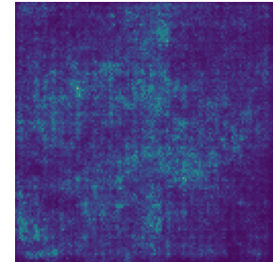
Raw Gradient



SmoothGrad



VarGrad



From the gif we can see that the images that were selected and identified as cats and dogs the image initially gets identified as dog due to the presents of other people, but as time goes it shifter to a classifier of cat and the gradients can be seen focusing more on the section with the cat in it. Then as the dog comes more into focus the classifier switches to dog and we see the gradients start to light up more focusing alot on the facial features. We can see in smoothgrad what the outline silhouette is of and how it changes through time, as both images are half visible we can see the gradients get very confused and not know what to focus on.